

# Notations

## 1. Greek Alphabet

| Uppercase  | Lowercase     | Name    |
|------------|---------------|---------|
| A          | $\alpha$      | Alpha   |
| B          | $\beta$       | Beta    |
| $\Gamma$   | $\gamma$      | Gamma   |
| $\Delta$   | $\delta$      | Delta   |
| E          | $\varepsilon$ | Epsilon |
| Z          | $\zeta$       | Zeta    |
| H          | $\eta$        | Eta     |
| $\Theta$   | $\theta$      | Theta   |
| I          | $\iota$       | Iota    |
| K          | $\kappa$      | Kappa   |
| $\Lambda$  | $\lambda$     | Lambda  |
| M          | $\mu$         | Mu      |
| N          | $\nu$         | Nu      |
| $\Xi$      | $\xi$         | Xi      |
| O          | $\omicron$    | Omicron |
| $\Pi$      | $\pi$         | Pi      |
| P          | $\rho$        | Rho     |
| $\Sigma$   | $\sigma$      | Sigma   |
| T          | $\tau$        | Tau     |
| $\Upsilon$ | $\upsilon$    | Upsilon |
| $\Phi$     | $\varphi$     | Phi     |
| $\chi$     | $\chi$        | Chi     |
| $\Psi$     | $\psi$        | Psi     |
| $\Omega$   | $\omega$      | Omega   |

## 2. Logical & Set Notation

| Symbol            | Name                       | Read As                 | Example                                                         |
|-------------------|----------------------------|-------------------------|-----------------------------------------------------------------|
| $\neg$            | Negation                   | Not                     | $\neg P$ (Not $P$ )                                             |
| $\wedge$          | Conjunction                | And                     | $P \wedge Q$ ( $P$ and $Q$ )                                    |
| $\vee$            | Disjunction                | Or                      | $P \vee Q$ ( $P$ or $Q$ )                                       |
| $\oplus$          | Exclusive Or               | Either... or            | $P \oplus Q$ (Either $P$ or $Q$ , but not both)                 |
| $\rightarrow$     | Implication                | If... then / Implies    | $P \rightarrow Q$ (If $P$ , then $Q$ )                          |
| $\leftrightarrow$ | Biconditional              | If and only if          | $P \leftrightarrow Q$ ( $P$ if and only if $Q$ )                |
| $\forall$         | Universal Quantifier       | For all / For every     | $\forall x P(x)$ (For all $x$ , $P(x)$ holds)                   |
| $\exists$         | Existential Quantifier     | There exists / For some | $\exists x P(x)$ (There exists an $x$ such that $P(x)$ )        |
| $\nexists$        | Non-existential Quantifier | There does not exist    | $\nexists x P(x)$ (No $x$ satisfies $P(x)$ )                    |
| $\exists!$        | Uniqueness Quantifier      | There exists a unique   | $\exists! x P(x)$ (There is exactly one $x$ where $P(x)$ holds) |
| $\top$            | Tautology                  | Top / Always true       | $P \vee \neg P \equiv \top$                                     |
| $\perp$           | Contradiction              | Bottom / Always false   | $P \wedge \neg P \equiv \perp$                                  |
| $\vdash$          | Syntactic Turnstile        | Proves / Yields         | $\Gamma \vdash P$ ( $\Gamma$ proves $P$ )                       |
| $\nvdash$         | Non-provability            | Does not prove          | $\Gamma \nvdash P$ ( $\Gamma$ cannot prove $P$ )                |
| $\models$         | Semantic Turnstile         | Entails / Satisfies     | $\Gamma \models P$ ( $\Gamma$ semantically entails $P$ )        |

| Symbol        | Name                | Read As                     | Example                                                 |
|---------------|---------------------|-----------------------------|---------------------------------------------------------|
| $\not\models$ | Non-entailment      | Does not entail             | $\Gamma \not\models P$ ( $\Gamma$ does not entail $P$ ) |
| $\equiv$      | Logical Equivalence | Is logically equivalent to  | $P \rightarrow Q \equiv \neg P \vee Q$                  |
| $\in$         | Element of          | Belongs to                  | $x \in A$ ( $x$ belongs to set $A$ )                    |
| $\notin$      | Not Element of      | Does not belong to          | $x \notin A$ ( $x$ is not in set $A$ )                  |
| $\subseteq$   | Subset              | Is a subset of              | $A \subseteq B$ ( $A$ is contained in or equal to $B$ ) |
| $\subset$     | Strict Subset       | Is a proper subset of       | $A \subset B$ ( $A$ is strictly contained in $B$ )      |
| $\not\subset$ | Not a Subset        | Is not a subset of          | $A \not\subset B$ ( $A$ is not a subset of $B$ )        |
| $\emptyset$   | Empty Set           | Empty set / Null set        | $A = \emptyset$ (Set $A$ has no elements)               |
| $\cup$        | Union               | Union of                    | $A \cup B$ (Elements in $A$ or $B$ or both)             |
| $\cap$        | Intersection        | Intersection of             | $A \cap B$ (Elements in both $A$ and $B$ )              |
| $\setminus$   | Set Difference      | Minus / Relative complement | $A \setminus B$ (Elements in $A$ but not in $B$ )       |

Ex.

$$\forall x(x \in A \cup B \rightarrow (x \in A \vee x \in B)) \wedge \nexists y(y \in \emptyset \wedge y \notin (A \setminus B))$$

Read as: “For all  $x$ ,  $x$  being an element of  $A$  union  $B$  implies that  $x$  is in  $A$  or  $x$  is in  $B$ ; and there exists no  $y$  such that  $y$  is in the empty set and  $y$  is not in  $A$  minus  $B$ .”

Ex.

$$(\Gamma \models P \oplus Q) \equiv (\Gamma \vdash P \vee Q) \wedge (\Gamma \not\vdash P \wedge Q) \wedge (P \wedge \neg P \equiv \perp)$$

Read as: “Gamma semantically entails P exclusive-or Q is logically equivalent to saying: Gamma proves P or Q, Gamma does not prove P and Q, and P and not P is a contradiction.”

Ex.

$$(\forall \alpha \in A, \exists! \beta \in B : f(\alpha) = \beta) \rightarrow (A \subseteq B \wedge A \not\subseteq \emptyset)$$

Read as: “If for every alpha in A there exists a unique beta in B such that f of alpha equals beta, then A is a subset of B and A is not a subset of the empty set.”

### 3. Mathematical Terms

- Definition: An explicit statement of the precise meaning of a mathematical term, symbol or proof.
- Theorem: A mathematical statement that has been proven to be true.
- Axiom / Postulate: A self evident truth or statement accepted as true without proof.
- Proposition: A standalone mathematical statement that is proven using axioms.
- Lemma: A minor or subsidiary theorem used as stepping stone to prove a larger theorem.
- Corollary: A proposition or statement that follows directly and easily from a previously proven theorem.
- Hypothesis: A proposed explanation or conditional assumption used as a basis of further reasoning or proof.
- Conjecture: A statement or mathematical hypothesis that is suspected to be true based on preliminary evidence

or intuition but has not yet been formally proved or disproved.